TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	57.7 ± 3.44
Correlation Stability	70.4 ± 1.96
Distribution Consistency	60.0 ± 2.53
Entropy Preservation	45.8 ± 3.63
Mean Z Score Drift	1.2 ± 0.15
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	-22.6 ± 15.72
Novelty Score	0.9 ± 0.01

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	47.925	-3.558	6.9%	OK
bp	76.469	79.500	+3.031	4.0%	OK
sg	1.017	1.013	-0.004	0.4%	OK
al	1.017	2.728	+1.711	168.3%	Drift
su	0.450	0.603	+0.153	34.0%	Drift
rbc	0.190	0.584	+0.395	208.4%	Drift
pc	0.227	0.562	+0.336	147.9%	Drift
pcc	0.106	0.259	+0.153	144.6%	Drift
ba	0.056	0.219	+0.163	293.8%	Drift
bgr	148.037	329.125	+181.088	122.3%	Drift
bu	57.426	157.314	+99.888	173.9%	Drift
sc	3.072	25.915	+22.843	743.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	57.7 ± 3.44
Correlation Stability (*2)	70.4 ± 1.96
Distribution Consistency (*3)	60.0 ± 2.53
Entropy Preservation (*4)	45.8 ± 3.63
Mean Z Score Drift (*5)	1.2 ± 0.15
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	-22.6 ± 15.72
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.3975	0.5387500000000001	0.5425	0.14500000000000002	*
F1	N/A	0.5440467505536792	0.469122235634558	0.469122235634558	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.7825	0.8724999999999999	-0.07999999999999999	*
F1	0.9587663012688215	0.8344984623971575	0.8769239250944093	-0.08184237617441227	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.7925	0.9775	-0.001250000000000067	
F1	0.9805471229310466	0.8522257903472796	0.9801612025017565	-0.00038592042928992 1	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9974999999999999	0.8225	0.9949999999999999	-0.002499999999999991	
F1	0.9978260869565216	0.8763484116391365	0.9956412949795302	-0.0021847919769914627	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.58375	0.53	-0.08249999999999999	*
F1	0.5885198185573248	0.7222541017286019	0.4512636888571197 6	-0.13725612970020507	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.9825000000000002	0.8625	0.9649999999999999	-0.017500000000000016	
F1	0.9848619694663338	0.8976504312546443	0.9681966994956774	-0.016665269970656583	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.945	0.8737499999999999	0.9649999999999999	0.020000000000000007	
F1	0.9465566844863709	0.8989873647151736	0.9688434345762038	0.022286750089832986	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9087500000000001	0.6425	0.8475000000000001	-0.061250000000000006	*
F1	0.9227021368025241	0.6932810398780889	0.8647235952135557	-0.05797854158896859	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.5912499999999999	0.6	-0.002500000000000022	
F1	0.7507214882964476	0.7427524788885893	0.7490410807976291	-0.0016804074988184703	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99	0.845	0.97625	-0.013749999999999995	
F1	0.9913018534863195	0.8721459169108569	0.978245097161552	-0.013056756324767604	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	90.0	1.015	4.0	0.0	0.0	1.0	0.0	0.0	490.0
2	2.0	90.0	1.015	4.0	0.0	1.0	1.0	0.0	0.0	490.0
3	90.0	80.0	1.01	3.0	3.0	1.0	0.0	1.0	1.0	22.0
4	90.0	90.0	1.015	3.0	0.0	1.0	1.0	0.0	0.0	490.0
5	90.0	80.0	1.01	3.0	3.0	1.0	0.0	1.0	1.0	22.0
6	90.0	90.0	1.015	3.0	0.0	1.0	1.0	0.0	1.0	490.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	90.0	1.01	4.0	1.0	0.0	1.0	1.0	0.0	22.0
2	90.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	490.0
3	90.0	60.0	1.01	2.0	0.0	0.0	0.0	1.0	0.0	22.0
4	90.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	490.0
5	90.0	80.0	1.01	2.0	0.0	0.0	0.0	1.0	0.0	490.0
6	2.0	70.0	1.01	2.0	1.0	0.0	1.0	1.0	0.0	22.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	60.0	1.01	3.0	0.0	0.0	0.0	1.0	0.0	490.0
2	2.0	90.0	1.01	2.0	0.0	0.0	1.0	1.0	1.0	22.0
3	90.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	22.0
4	90.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	490.0
5	2.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	22.0
6	90.0	60.0	1.01730902777779	0.0	0.0	0.0	0.0	0.0	0.0	490.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	70.0	1.01	3.0	0.0	0.0	1.0	0.0	1.0	490.0
2	90.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	22.0
3	2.0	60.0	1.015	3.0	0.0	1.0	0.0	0.0	0.0	490.0
4	2.0	80.0	1.01	3.0	0.0	0.0	1.0	0.0	0.0	490.0
5	2.0	80.0	1.01	3.0	0.0	0.0	1.0	0.0	0.0	490.0
6	90.0	70.0	1.015	0.0	0.0	1.0	1.0	0.0	0.0	22.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	90.0	1.01	3.0	0.0	0.0	1.0	1.0	0.0	22.0
2	2.0	90.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	22.0
3	90.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	490.0
4	90.0	60.0	1.01	3.0	0.0	0.0	0.0	0.0	1.0	490.0
5	90.0	80.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	490.0
6	90.0	90.0	1.01	3.0	0.0	0.0	0.0	1.0	1.0	490.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	490.0
2	2.0	70.0	1.02	4.0	0.0	1.0	0.0	0.0	0.0	22.0
3	90.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	22.0
4	90.0	70.0	1.02	4.0	0.0	0.0	0.0	0.0	0.0	22.0
5	2.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	22.0
6	90.0	90.0	1.01	4.0	0.0	0.0	1.0	1.0	0.0	22.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	3.0	70.0	1.02	4.0	0.0	0.0	1.0	0.0	0.0	490.0
2	3.0	60.0	1.015	4.0	2.0	0.0	1.0	1.0	0.0	490.0
3	90.0	70.0	1.015	3.0	0.0	0.0	0.0	0.0	0.0	22.0
4	3.0	60.0	1.015	3.0	2.0	0.0	1.0	0.0	0.0	490.0
5	90.0	70.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	22.0
6	3.0	70.0	1.005	3.0	0.0	0.0	1.0	0.0	0.0	22.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	70.0	1.01	0.0	0.0	0.0	1.0	0.0	0.0	22.0
2	2.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	490.0
3	90.0	80.0	1.01	4.0	0.0	0.0	1.0	0.0	0.0	490.0
4	90.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	490.0
5	90.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	490.0
6	2.0	70.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	22.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	2.0	90.0	1.015	4.0	0.0	0.0	0.0	0.0	0.0	463.0
2	2.0	70.0	1.015	3.0	0.0	0.0	0.0	0.0	0.0	22.0
3	90.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	22.0
4	90.0	70.0	1.02	1.0	0.0	0.0	0.0	0.0	0.0	463.0
5	90.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	463.0
6	90.0	70.0	1.02	1.0	0.0	0.0	0.0	0.0	0.0	463.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	90.0	60.0	1.01	4.0	1.0	1.0	1.0	1.0	0.0	490.0
2	2.0	60.0	1.01	4.0	1.0	1.0	1.0	0.0	1.0	490.0
3	90.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	22.0
4	90.0	70.0	1.01717081850533	0.0	0.0	0.0	0.0	0.0	0.0	490.0
			8							
5	2.0	70.0	1.015	4.0	0.0	0.0	1.0	0.0	0.0	490.0
6	90.0	80.0	1.01717081850533	0.0	0.0	0.0	0.0	0.0	0.0	490.0
			8							
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0